

UNIT 3

1. Fundamentals of Angular JS

AngularJS is a JavaScript-based open-source front-end framework maintained by Google. It is used for building dynamic, single-page web applications (SPAs) and provides a structured way to build web apps by extending HTML capabilities.

General Features

The general features of AngularJS are as follows –

- AngularJS is a efficient framework that can create Rich Internet Applications (RIA).
- AngularJS provides developers an options to write client side applications using JavaScript in a clean Model View Controller (MVC) way.
- Applications written in AngularJS are cross-browser compliant. AngularJS automatically handles JavaScript code suitable for each browser.
- AngularJS is open source, completely free, and used by thousands of developers around the world. It is licensed under the Apache license version 2.0.

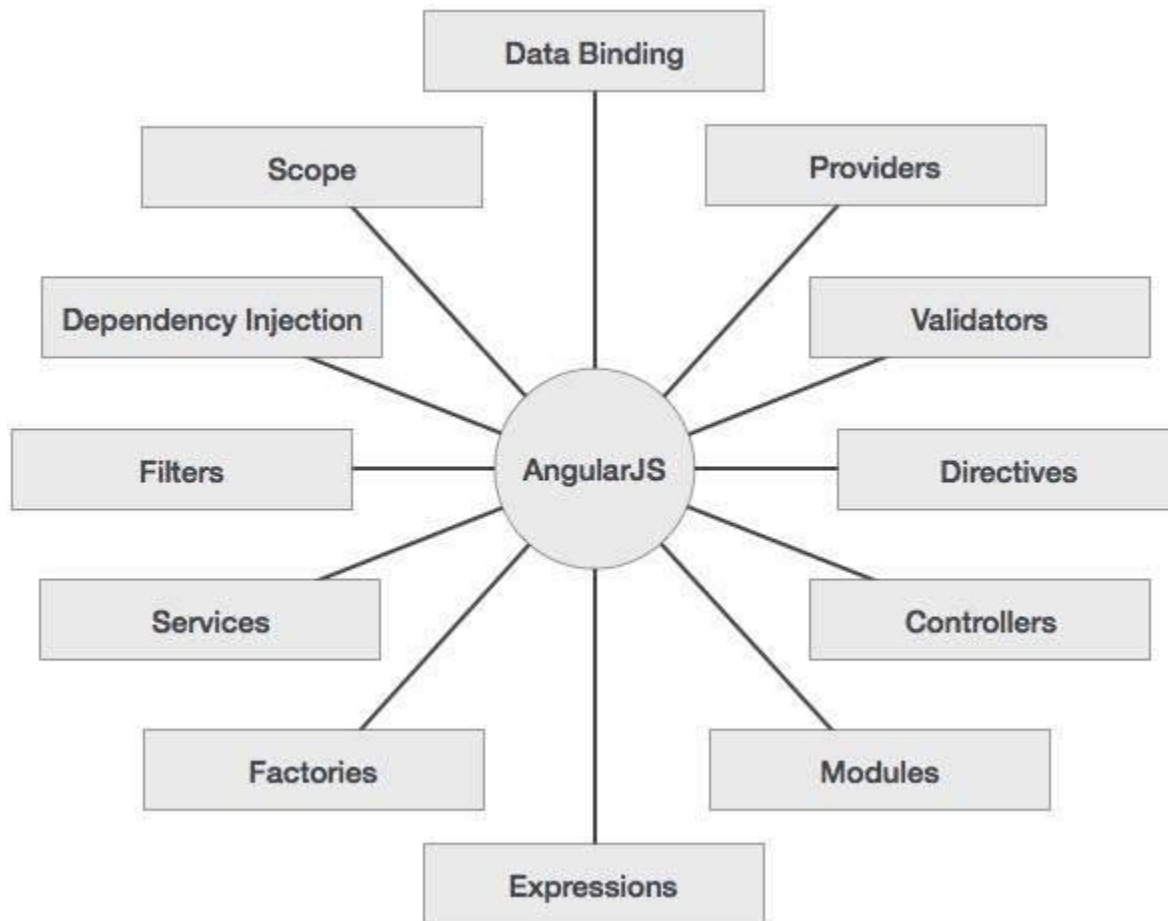
Core Features

The core features of AngularJS are as follows –

- **Data-binding** – It is the automatic synchronization of data between model and view components.
- **Scope** – These are objects that refer to the model. They act as a glue between controller and view.
- **Controller** – These are JavaScript functions bound to a particular scope.
- **Services** – AngularJS comes with several built-in services such as \$http to make a XMLHttpRequests. These are singleton objects which are instantiated only once in app.
- **Filters** – These select a subset of items from an array and returns a new array.
- **Directives** – Directives are markers on DOM elements such as elements, attributes, css, and more. These can be used to create custom HTML tags that serve as new, custom widgets. AngularJS has built-in directives such as ngBind, ngModel, etc.
- **Templates** – These are the rendered view with information from the controller and model. These can be a single file (such as index.html) or multiple views in one page using *partials*.
- **Routing** – It is concept of switching views.
- **Model View Whatever** – MVW is a design pattern for dividing an application into different parts called Model, View, and Controller, each with distinct responsibilities. AngularJS does not implement MVC in the traditional sense, but rather something closer to MVVM (Model-View-ViewModel). The Angular JS team refers it humorously as Model View Whatever.
- **Deep Linking** – Deep linking allows to encode the state of application in the URL so that it can be bookmarked. The application can then be restored from the URL to the same state.
- **Dependency Injection** – AngularJS has a built-in dependency injection subsystem that helps the developer to create, understand, and test the applications easily.

Concepts

The following diagram depicts some important parts of AngularJS which we will discuss in detail in the subsequent chapters.



Advantages of AngularJS

The advantages of AngularJS are –

- It provides the capability to create Single Page Application in a very clean and maintainable way.
- It provides data binding capability to HTML. Thus, it gives user a rich and responsive experience.
- AngularJS code is unit testable.
- AngularJS uses dependency injection and make use of separation of concerns.
- AngularJS provides reusable components.
- With AngularJS, the developers can achieve more functionality with short code.
- In AngularJS, views are pure html pages, and controllers written in JavaScript do the business processing.

On the top of everything, AngularJS applications can run on all major browsers and smart phones, including Android and iOS based phones/tablets.

Disadvantages of AngularJS

Though AngularJS comes with a lot of merits, here are some points of concern –

- **Not Secure** – Being JavaScript only framework, application written in AngularJS are not safe. Server side authentication and authorization is must to keep an application secure.
- **Not degradable** – If the user of your application disables JavaScript, then nothing would be visible, except the basic page.

1. AngularJS Expressions

- **Definition:** AngularJS expressions are used to bind data to HTML. They are written inside curly braces `{{ expression }}`.
- **Features:**
 - Can contain literals, operators, and variables.
 - Are evaluated against the AngularJS `$scope`.
 - Do not use a control structure (like loops or conditionals).
 - Automatically update when the scope variables change.
- **Example:**

html

Copy code

```
<div ng-app="myApp" ng-controller="myCtrl">
  {{ 5 + 5 }} <!-- Output: 10 -->
  {{ firstName + " " + lastName }} <!-- Concatenates strings -->
</div>
<script>
  var app = angular.module('myApp', []);
  app.controller('myCtrl', function($scope) {
    $scope.firstName = "John";
    $scope.lastName = "Doe";
  });
</script>
```

2. AngularJS Arrays

- **Definition:** Arrays in AngularJS are JavaScript arrays used within `$scope` objects.
- **Use Cases:** Can be used to display lists using `ng-repeat`.
- **Example:**

html

Copy code

```
<div ng-app="myApp" ng-controller="myCtrl">
  <ul>
    <li ng-repeat="fruit in fruits">{{ fruit }}</li>
  </ul>
</div>
<script>
  var app = angular.module('myApp', []);
```

```
app.controller('myCtrl', function($scope) {
  $scope.fruits = ["Apple", "Banana", "Cherry"];
});
</script>
```

3. AngularJS Objects

- **Definition:** Objects in AngularJS are similar to JavaScript objects, consisting of key-value pairs, and are used within \$scope to hold data.
- **Example:**

```
html
Copy code
<div ng-app="myApp" ng-controller="myCtrl">
  {{ person.firstName + " " + person.lastName }}
</div>
<script>
  var app = angular.module('myApp', []);
  app.controller('myCtrl', function($scope) {
    $scope.person = { firstName: "John", lastName: "Doe" };
  });
</script>
```

4. AngularJS Strings

- **Definition:** Strings in AngularJS are treated the same as JavaScript strings.
- **Binding Strings:** You can concatenate or manipulate strings using AngularJS expressions or controller logic.
- **Example:**

```
html
Copy code
<div ng-app="myApp" ng-controller="myCtrl">
  {{ greeting }}
</div>
<script>
  var app = angular.module('myApp', []);
  app.controller('myCtrl', function($scope) {
    $scope.greeting = "Hello, " + "World!";
  });
</script>
```

5. AngularJS Form Validation & Form Submission

- **Definition:** AngularJS provides built-in validation for form fields such as required, minlength, maxlength, email, etc., and allows custom validations.
- **Directives:**
 - ng-model: Binds input fields to scope variables.
 - ng-submit: Defines an action when the form is submitted.

- Validation classes: ng-valid, ng-invalid, ng-dirty, ng-pristine.

- **Example:**

html

Copy code

```
<div ng-app="myApp" ng-controller="myCtrl">
  <form name="myForm" ng-submit="submitForm()">
    <input type="text" name="username" ng-model="username" required>
    <span ng-show="myForm.username.$error.required">Username is
required.</span>
    <button type="submit">Submit</button>
  </form>
</div>
<script>
  var app = angular.module('myApp', []);
  app.controller('myCtrl', function($scope) {
    $scope.submitForm = function() {
      alert("Form submitted with username: " + $scope.username);
    };
  });
</script>
```